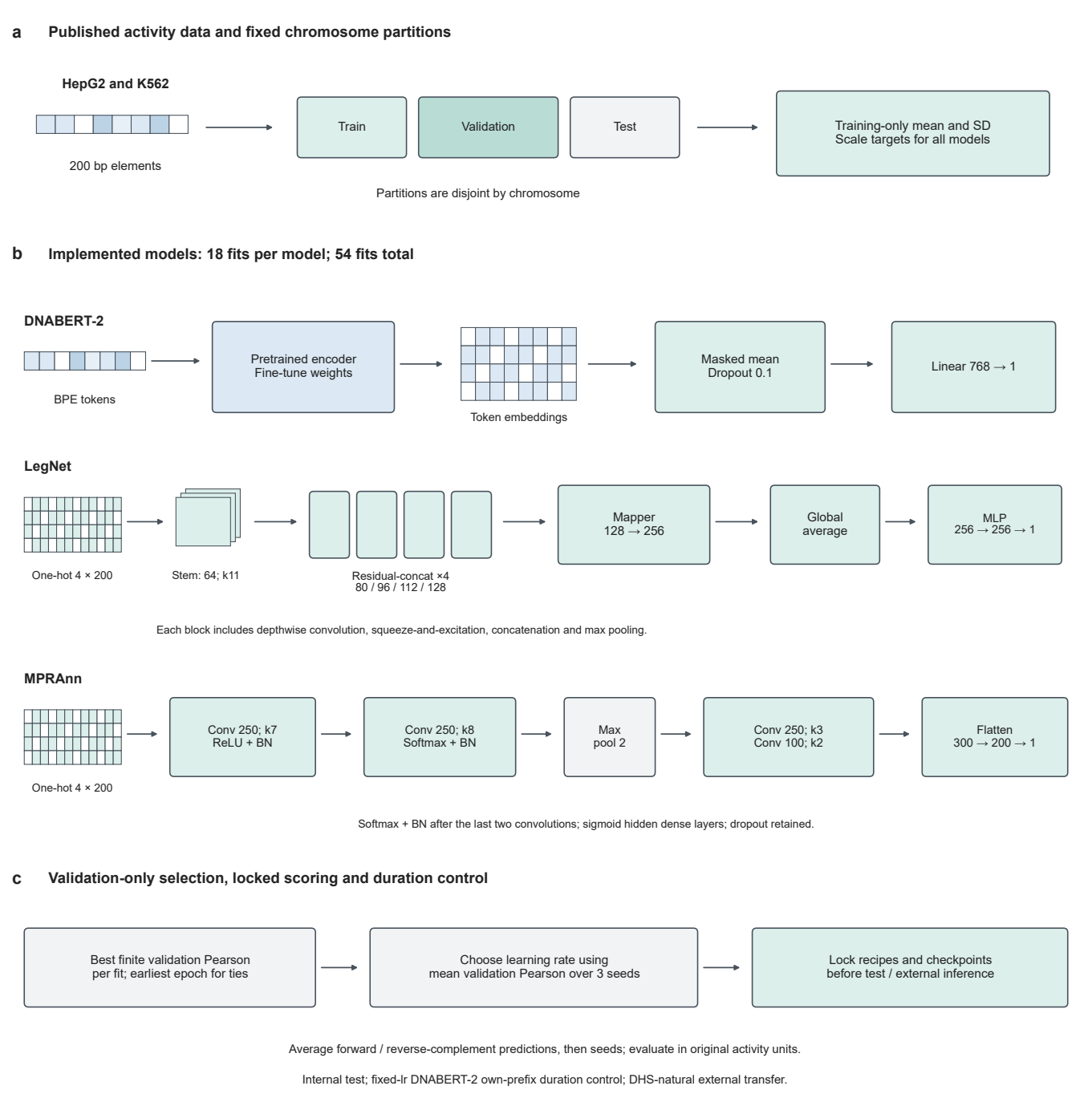

a
Published activity data and fixed chromosome partitions
HepG2 and K562
200 bp elements
Train
Validation
Test
Training-only mean and SD
Scale targets for all models
Partitions are disjoint by chromosome
b
Implemented models: 18 fits per model; 54 fits total
DNABERT-2
BPE tokens
Pretrained encoder
Fine-tune weights
Token embeddings
Masked mean
Dropout 0.1
Linear 768 → 1
LegNet
One-hot 4 × 200
Stem: 64; k11
Residual-concat ×4
80 / 96 / 112 / 128
Mapper
128 → 256
Global
average
MLP
256 → 256 → 1
Each block includes depthwise convolution, squeeze-and-excitation, concatenation and max pooling.
MPRAnn
One-hot 4 × 200
Conv 250; k7
ReLU + BN
Conv 250; k8
Softmax + BN
Max
pool 2
Conv 250; k3
Conv 100; k2
Flatten
300 → 200 → 1
Softmax + BN after the last two convolutions; sigmoid hidden dense layers; dropout retained.
c
Validation-only selection, locked scoring and duration control
Best finite validation Pearson
per fit; earliest epoch for ties
Choose learning rate using
mean validation Pearson over 3 seeds
Lock recipes and checkpoints
before test / external inference
Average forward / reverse-complement predictions, then seeds; evaluate in original activity units.
Internal test; fixed-lr DNABERT-2 own-prefix duration control; DHS-natural external transfer.